History & Development of medical microbiology

Microbiology – III

BVOC – HCT

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History of Medical Microbiology

- The medical microbiology had its origin in the year 1888 when the Pasteur Institute in Paris (France) was established.
- Robert Koch became the Director of the Institute For Infective Diseases.
- Both Louis Pasteur and Robert Koch infused the spirit and interest to gain knowledge in medical microbiology to the countrymen of America and Europe.
- The medical microbiology deals with the aspects of infection, the causative agents of infection and the diseases due to infection.
- Following the establishment of Germ Theory of Diseases by Pasteur, innumerable infectious diseases that haunt the human population were discovered.
- For most of these infectious diseases, the drugs and vaccines have been found.

Contributions of Nobel laureates

Year	Nobel Laureate	Contribution
1902	Ronald Ross	Malaria
1905	Robert Koch	Tuberculosis
1908	Paul Ehrlich	Theories on immunity
1908	Elie Metchnikoff	Phagocytes
1928	Nicolle	Typhus
1939	Domagk	Prontosil as antibacterial
1945	Alexander Fleming, Chain & Florey	Penicillin
1952	Selman A. Waksman	Streptomycin
1954	John Franklin Enders	Poliomyelitis
1966	Francis Peyton Rous	Rous sarcoma virus

Sir Ronald Ross

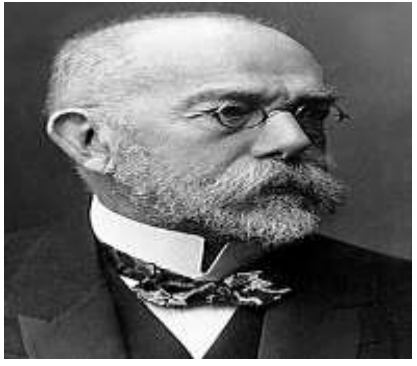
Nobel Prize for Physiology or Medicine (1902)



- He was a British medical doctor
- Received the Nobel Prize for Physiology or Medicine in 1902 for his work on the **transmission of malaria.**
- Became the first British Nobel laureate, and the first born outside Europe.
- His discovery of the malarial parasite in the gastrointestinal tract of a mosquito in 1897
  - Proved that malaria was transmitted by **mosquitoes**
  - Laid the foundation for the method of combating the disease.
- Malarial parasite were observed in the Anapheles mosquitoes and called it “Dapple wings”
- Sick birds were used to study lifecycle of the malarial parasite

- Ronald Ross was awarded a Nobel Prize for his discovery of the **life cycle of malarial parasite** (*Plasmodium* sp.) in birds.
- He did not build his concept of malarial transmission in humans, but in birds.
- Ross was the first to show that malarial parasite was transmitted by the bite of infected mosquitoes.
- Ross developed mathematical models for the study of malaria epidemiology, which he initiated in his report on Mauritius in 1908.
- He elaborated the concept in his book The Prevention of malaria in 1910.

Robert Koch (1905)



- German physician & microbiologist
- **Father of modern bacteriology**
- Robert Koch, a younger contemporary of Pasteur, made his contributions on new procedures for staining, visualizing and growing bacteria.
- He solidified liquid culture media with agar.
- He also isolated and characterized the bacilli of **anthrax**, and **tuberculosis** and demonstrated their causative role by a series of experiments.
- After identifying the tubercle bacillus, Koch has advocated his postulate, in order to distinguish a pathogenic microbe from adventitious microbes.

- The impact of his work towards the end of 19th century lead to the discovery of
  - 'specific toxins' made by the causative bacteria for **diphtheria** and **tetanus**
  - 'antibodies' produced in host animals against the toxins to neutralise their effect.
- His finding also lead to immunization therapy.
- He was awarded the Nobel Prize in the year 1905 for his work on **tuberculosis**.

Paul Ehrlich
(1908)



- German physician and scientist
- Worked in the fields of Hematology, immunology, and antimicrobial **chemotherapy**.
- Among his foremost achievements were finding a cure for **syphilis** in 1909
- Inventing the precursor technique to **Gram staining bacteria**.
- His laboratory discovered arsphenamine (**Salvarsan**), the first effective medicinal treatment for syphilis, thereby initiating and also naming the concept of chemotherapy.
- Ehrlich popularized the concept of a **magic bullet**, chemical substances which have special affinities for pathogenic organisms.

- He produced **trypan red** dye effective against trypanosomes.
- He developed a new staining technique to identify the tuberculosis bacillus that had been discovered by the German bacteriologist Robert Koch.
- Ehrlich also differentiated the numerous types of blood cells of the body and thereby laid the foundation for the field of hematology.

- He also made a decisive contribution to the development of an **antiserum** to combat **diphtheria** and conceived a method for standardizing therapeutic serums.
- In 1908, he received the Nobel Prize in Physiology or Medicine for his contributions to **immunology**.
- He was the founder and first director of what is now known as the Paul Ehrlich Institute, a German research institution and medical regulatory body that is the nation's federal institute for vaccines and biomedicines.
- A genus of *Rickettsiales* bacteria, *Ehrlichia*, is named after him.

Élie Metchnikoff



- Elie Metchnikoff was a Russian Imperial zoologist
- He is best known for his pioneering research in **immunology**
- He and Paul Ehrlich were jointly awarded the 1908 Nobel Prize in Physiology or Medicine "in recognition of their work on **immunity**“.
- Honoured as the "**father of innate immunity**,“
- He was the first to discover a process of immunity called **phagocytosis** and the cell responsible for it, called phagocyte, specifically macrophage, in 1882.
- This led to the foundation of innate immunity and cell mediated immunity.

Charles Nicolle



- Charles Jules Henri Nicolle (1866 – 1936) was a French bacteriologist .
- Received the Nobel Prize in Medicine for his identification of **lice** as the transmitter of **epidemic typhus**.
- He also showed that the major transmission method was not louse bites but excrement that contained large number of microbes.
- Attempted to make vaccines using sodium fluoride, as a good reagent to sterilize parasites and preserves their structures.
- Using this method, he developed **vaccines** for gonorrhea, some staphylococcal infections, and cholera.

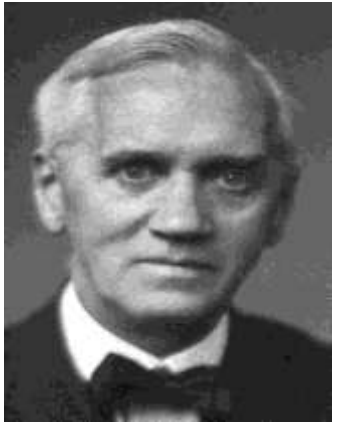
Gerhard Domagk



- Gerhard Johannes Paul Domagk was a German pathologist and bacteriologist.
- Discovered of Sulfonamidochrysoidine (KI-730)
- First commercially available **antibiotic** and marketed under the brand name **Prontosil**.
- He received the 1939 Nobel Prize in Physiology or Medicine.
- Domagk's work on **sulfonamides** eventually led to the development of the antituberculosis drugs thiosemicarbazone and isoniazid,
- which helped to curb the epidemic of tuberculosis which swept Europe after World War II.

Alexander Fleming, Chain & Florey

- Fleming (London University), Chain and Florey (Oxford University)
- Awarded the 1945 Nobel Prize in Physiology or Medicine
- For their discovery of the antibiotic **penicillin** against bacterial diseases.
- Fleming noticed that some bacteria were killed and had dissolved away around a spot of blue-green mould which by chance had contaminated one of his dishes.
- This mould belonged to the *Penicillium* species and named the antibacterial substance as penicillin.
- Chain and Florey went on to **purify** and **extract penicillin**,
- Enabling it to be produced in large amounts to treat many different bacterial diseases.



Alexander Fleming



Penicillium sp.

Selman A. Waksman



- Russian Empire-born Jewish-American inventor, biochemist and microbiologist.
- Studied the **decomposition** of organisms that live in **soil**
- This led to the discovery of **streptomycin** and several other antibiotics.
- In 1952, he was awarded the Nobel Prize in Physiology or Medicine for "ingenious, systematic and successful studies of the soil microbes that led to the discovery of streptomycin."

John Franklin Enders



- American biomedical scientist and Nobel Laureate.
- Enders has been called "**The Father of Modern Vaccines**".

- In 1949, Enders, Thomas Huckle Weller, and Frederick Chapman Robbins reported successful ***in vitro*** **culture** of an animal virus—**poliovirus.**

- They received the 1954 Nobel Prize in Physiology or Medicine "for their discovery of the ability of poliomyelitis viruses to grow in cultures of various types of tissue"

Francis Peyton Rous



- American Nobel Prize-winning virologist.
- Rous was involved in the discovery of the role of **viruses** in the transmission of certain types of **cancer**.
- In 1966, he was awarded a Nobel Prize in Physiology or Medicine for his work.
- In 1911, as a pathologist he found that cancer could be transmitted by a virus - known as **Rous sarcoma virus**.
- Received Nobel Prize for Medicine or Physiology after 40 years of this finding.